Lesson 5-1 · Single-Period Quick

nth Roots, Rational Exponents + Cylinder DOK-3 (Item 1A Prep)

Klimsara-adapted · Student-centered · No Blooket

Do Now — $y = x^2$ graph: roots from squares

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DOK 1 | 5 min | *bridge to nth roots*

Do Now — Explore & Reason: $y = x^2$Graph of $y = x^2$ is projected. For each, find the missing value:

(a) $(2,\ \_\_)$ (b) $(3,\ \_\_)$ (c) $(-3,\ \_\_)$ (d) $(5,\ \_\_)$

(e) $(\_\_,\ 4)$ (f) $(\_\_,\ -16)$ (g) $(\_\_,\ 7)$

B. How do you find $\sqrt{13}$ using this graph?

C. Sketch $y = x^3$. Approximate $\sqrt[3]{5}$, $\sqrt[3]{-5}$, $\sqrt[3]{8}$.

Launch — Example 1 + Example 3 (two examples, compressed)

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DOK 1–2 | 13 min | *teacher models*

nth Roots ↔ Rational Exponents — Rules Card

1. $x^{1/n} = \sqrt[n]{x}$ (principal nth root)
2. $x^{m/n} = \left(\sqrt[n]{x}\right)^m = \sqrt[n]{x^m}$
3. Rationalize nth-root denominator: multiply by $\sqrt[n]{x^{n-1}}$
4. Even index: radicand ≥ 0 in Reals; odd index: any real

Example 1: real nth roots — **Example 3:** evaluate rational exponents

Release to Explore at minute 18. Copy Rules card into margin.

Explore — TPS items 1–4 (first 14 min)

DOK 2–3 | 32 min | *student work*

7 items in order. Plan ~10 min for Practice #50.

Try It 1 · Real nth roots Find the real fourth roots of 81; the real cube roots of 64.

Try It 3 · Evaluate rational exponents Evaluate $-(16^{3/4})$ and $\sqrt[5]{3.5^4}$. Round if needed.

Practice #28 · nth roots with variables Find the real square roots of 25.

Practice #30 · Rational exponent form Rewrite $\sqrt[6]{729}$ using a fractional exponent.

Explore — TPS items 5–7 + DOK-3 Capstone (18 min)

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DOK 2–3 | 32 min | *student work*

Practice #37 · Simplify: mixed radical/rational Simplify $\sqrt[6]{729a^{24}b^{18}}$. Assume all variables positive.

Practice #48 · Multi-step rational exponent Which expressions equal $b^{3/4}$? (Yes/No table — see packet.)

Practice #50 · CYLINDER MODELING ★ Cylindrical container: height = diameter. Volume = 169.65 ft^3.

Part A: lateral surface area? **Part B:** containers in 20-ft hold?

Set up $V = \pi r^2(2r)$, isolate r in rational-exponent form, then rationalize.

Share / Summary — Cylinder Reveal

DOK 2 **5 min** *DOK-3 reveal + assessment bridge*

Sentence frame · Cylinder Capstone "I isolated r from $V = \pi r^2 h$ by substituting $h = 2r$, giving $V =$ __________. Solving for r, I got $r =$ __________. In rational-exponent form, $r = \left(V/(2\pi)\right)^{1/3}$. To rationalize I multiplied by ______ to get __________."

Bridge to Topic 5 Assessment — Item 1A Prep The cylinder move: isolate a variable $\to$ rational-exponent form $\to$ rationalize. This is **exactly** what Topic 5 Assessment Item 1A asks. You have now rehearsed it.

Exit Ticket — Summary Recap

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DOK 1–2 | 3 min | *not graded*

Complete on your exit ticket

1. $x^{m/n}$ is the same as ______________________ in radical form.
2. To rationalize a cube-root denominator I __________________.
3. One biggest thing I learned today: __________________.